

ActInf GuestStream #022.1 ~ Anna Lembke, "Dopamine Nation"

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hello and welcome everyone this is active lab guest stream number 22.1 i'm here with dr anna lumke and we are here to discuss dopamine and beyond and really looking forward to this conversation and appreciate the time so to you for any introductory context and then really looking forward to this discussion yeah well thank you so much for having me daniel you know i've uh been uh you know interested in you and your work for many years now and so i'm really excited to have an excuse to reconnect um and just excited to be here and you asked me to start by introducing myself so i'm i'm a professor of psychiatry and behavioral sciences and also addiction medicine at lanford university school of medicine i'm also author of the book dopamine nation finding balance in the age of indulgence and that's what we're going to talk about which i and daniel's quoted in there and i would have actually included daniel moore in there if i had if i were smart enough to actually understand the work that he does thanks um we'll get there with the ants and the dopamine but yeah just what is dopamine nation about and what motivated you to write it so dopamination is kind of a blend of neuroscience made simple combined with practical advice for how to live in a dopamine overloaded world holding up people in recovery from severe addictions as modern day profits for the rest of us because all the neuroscience in the world as valuable as it is um can't make up for the whole experience of people who have gotten into recovery from addiction um dopamine nation also posits a kind of global hypothesis you know tentatively posits the hypothesis that the growing rates of anxiety depression suicide and other mental health disorders across the world are caused at least in part by the problem of overabundance uh and the kind of lack of appropriate friction in our lives in other words the ways in which our lives are maybe too easy and the way that our primitive wiring is mismatched for this modern ecosystem of overwhelming over abundance okay one dopamine question and then i think we'll start to see some areas of overlap with active inference what could you say to the role of dopamine in reward and motivation signaling like in wanting and liking and how does

dopamine play a role in behavioral selection okay so dopamine is essential for the experience of motivation reward and pleasure it may be even more important for motivation than it is uh for the experience of pleasure and i base this on the work of my amazing neuroscience colleagues who have done experiments for example where they engineer a rat's brain to do completed dopamine receptors and discover that if they give that rat food it will eat the food and seem to take pleasure in the food but if they put the food you know some distance away the rat will starve to death meaning that the absence of dopamine in these key reward circuits um make it such that that animal is no longer motivated to do the work that it takes uh to get their reward and my neuroscience colleagues like rob malenka had said that for him the definition of addiction is how hard an animal is willing to work to get their reward and that has direct application to the clinical work that i do um you know we all like what we like and once we experience the thing that we like of course we want to repeat it but the kind of fundamental difference between those who go on to develop addiction and those who don't is that those who develop addiction are willing to work very very hard to get their drug of choice and are willing to give up a lot of other things even when that pursuit causes enormous amount of you know personal costs thank you for that that's really powerful with professor malenka's definition of how hard they work to get it because it's like an action-oriented definition that could be measured in or proxied by different things especially clinically as opposed to like in inference or a purely cognitive based report like a survey how often do you think about something like that and uh that's very interesting and that's i think one of the themes that we'll pick up on which is action and for active inference it's right there in the title it's like there's always going to be a minimum of two and when we're talking about behavior and cognition they're an inseparable whole and then that plays out in a lot of different ways like partially with the embodiment and the inactive perspectives that come to view but also just like as you said it's pragmatic too because it's relevant to always be thinking about action and cognition as like a dyad in the context rather than trying to separate them and then find a correlation across love it love it so i've been a psychiatrist for uh going on 30 25 years and i would say the first half of my training and life experience and education was really focused on changing people's cognitions so focusing on people's thoughts and emotions and using language as a tool to talk about those to change those but as my career went on i became more and more skeptical of this as a solitary pursuit and more and more convinced that what we need to do is focus on changing behavior and when i'm dealing with patients with addiction that is in fact now my first intervention i will say let us first change the behavior and in doing that experiment and

what i mean is abstain from your drug of choice see what happens and then let's make certain inferences based on what that experience was like for you and that's really a typical in psychiatry today what we do instead is we talk and we talk and we talk about why the behaviors are happening without really focusing on changing the behaviors and i think we really that that's not good like because for example my patients will come in they'll be addicted and they'll say i'm addicted because i'm depressed or because i have this trauma all of that may be true but we can talk about the trauma till the cows come home you have to change the behavior and you have to do it before you feel like it so that's really interesting too like patients will say well you know i'm just not i don't i'm not motivated to do x y or z so this is an earlier like i don't feel i said you know what if you wait till you feel like doing it you will never do it what you need to do is change the behavior experience the impact on your feeling by changing the behavior and then take it from there so i totally agree with you it's this intense marriage of cognition emotion and the actual actions and we almost always are focused on you know the cognitions and emotions and we don't get people to do enough of just focusing on changing behavior okay so i think that's our first theme which is this action orientation and that plays into the embodiment and the inactive aspects but also it has a kind of corollary this like act first principle and that helps us perturb the causal structure because if you're just looking at the n equals one observational experiment well your consumption will be correlated with the price of oranges and then how do you actually get at something deeper in the world and that's sort of a relevant type of intervention so action and also action first is pretty cool any other thoughts on that or we can go in a different direction well i would also say that this brings to mind you know you asked about surveys and things like that like i don't really like just personally like i don't really i mean i think there's value for for like survey studies but the problem is what you really end up picking up there is more about what the dominant cultural narrative is that explains why people do x y or z and you're not really tapping into why they really do x y or z the other thing i would say that you know the way that we think about science is that it's not a single person's life narrative but i really believe that that that there's science when you see a series of individuals over and over again you hear their stories and you can recognize very similar patterns and then you do an intervention and you see the results and you do that repeatedly with many people over time it's not like your classic you know a clinical study but it feels really true and real to me in a way that a lot of times clinical studies don't or can't capture some of the deeply nuanced complex aspects of human behavior i love the way you think about things it's so crazy now you're like i didn't know what was gonna

come next so um yeah um all right so back to the book and also to connect this to an active inference theme so if decision making in active inference like policy selection the pi variable the way that different possibilities are ranked is not just by their reward or their expected like value or utility which would be common in a kind of reinforcement learning or a reward maximization framework but there's like two components that go into that policy selection there's an epistemic knowledge oriented like learning and curiosity intrinsic motivation and then there's a pragmatic so there is a utility element and the ability to dial in on utility but also there's the ability to kind of pull back and when you were describing your book structure certainly while reading it and as you described it there was like an epistemic component neuroscience made accessible for those who might not have the molecular background but also like on page 48 with a baseball game it's just fun for even those who might have seen it in a different way and then there's the pragmatic part which uses the stories um to kind of connect to useful tips so where do you think epistemic and pragmatic value come into play in how you're thinking about this um so so epidemic is interesting um and thank you for showing the examples related to my book um because that helps me also understand what you mean by those terms so what i find really interesting um about you know how to intrinsically motivate people with addiction is that if they have a framework in which to understand their behavior it really does help people change their it's really a framework and what's fascinating to me is that neuroscience talks and everything else walks and by that i mean when i explain people their behavior with a neuroscience framework it carries a lot of currency to people for people it feels real you know it feels like it's almost like let me just throw it out there like a modern day religion like you could use a very different framework like you could use a moral framework you could use a philosophical framework people don't resonate with that the same way they resonate with the neuroscience which is why i use the neuroscience framework um so i think and people will carry it around with them so for example i use this extended metaphor of a balance pleasure pain balance and gremlins represent neuro adaptation and they hop on the pain side of balance and for example i had a reader write me and say you know i'm trying really trying really hard to quit smoking and whenever i have an urge to smoke a cigarette i just imagine those little gremlins that was slowing their adaptation hopping up and down in the painting south i'm not you're not gonna win you know i'm going to outlast you um so that's really interesting people will you know it's like a frame we need those epistemological frameworks that we project onto the world to help us cope make decisions last you know in difficult situations nice thanks um maybe even one way that that person is

experiencing their experience the way that they're thinking about their behavior and inference it's um like the the metaphor that you're bringing the baseball game the for the neurotransmitter synapse the gremlins even some of the more flow charts like how different stages are related to each other there's always like a person's form moving through it and then that provides a lot of accessibility to enter into that space and like kind of imagine counterfactuals which is our human cognitive expertise in a sense is not to memorize 11 digit numbers even outside of ultra specific training it's to okay if the neurotransmitter like when i looked at that example i thought well where are the glia are they the crowd are they like what are they doing are they supporting this pitch and then there's that time scale but then it already opens the door to questions that the molecular representation's like oh well then it's a done deal right with the pre the post synaptic yeah right and i think you know obviously these metaphors are like a gross oversimplification and get things you know can't explain all of the phenomena but as you say if we try to explain it all and we get too complicated then it feels to people like memorizing a list of numbers so it has to be true enough to represent the real science but not so complex that people can't take it with them and apply it in their everyday everyday lives and it also it takes it out of the the kind of uncanny valley an example there would be like the ribosome it's a purple blob and right well it's really not but it's represented so commonly and then that locks in but then at the baseball game we know it's not it's a different scale so it's approachable and then anyone who wants to look into the receptor slash glove yes yeah so um um that was kind of exploring what was the second area which was like the epistemic and the pragmatic like always pairing the knowledge gaining with the useful and then recognizing that there's a time for the more knowledge oriented maybe there's a building on campus that's more knowledge oriented and then also there's a time and a place for what's useful and maybe that's part of the conversation that do you want to see or how do you want to see it with applied neuroscience can you can you say what you mean a little bit more i didn't quite get that like connecting the epistemic and the pragmatic the learning and research oriented work with the practical tips is that disconnect how could it be how should it be how will we make that connection like flexible and and work yeah i mean there's obviously you know a huge disconnect like this is this is the problem we're sort of you know in this ivory tower we're we're learning this stuff we're acquiring this knowledge we're working you know people you know neuroscientists like you actually adding new knowledge but yeah then the question is how do we get that to people to actually you know benefit them um you know in their everyday lives and also get it getting it to to students to give them a sense of meaning and

purpose in their work so it doesn't feel so incredibly disembodied from the world that we live in and the reasons behind why we do i mean why do we want to acquire this knowledge i mean you know there's there's not just that i mean maybe it is knowledge for its own sake but i think deeply people want to feel they're contributing making the world a better place uh you know so i think really important to to do we need to do more to bring those things together i feel like you know medical education the so there's the pre-clinical years and then there's the clinical years and i know for me personally like those clinical years were where i first sort of felt like oh thank goodness you know here i am interacting with like real human beings and trying to take this knowledge and also i'm impatient i'm very impatient so i feel like the human interaction i could see more quickly that i was making a difference i didn't have the kind of patience that it takes to be a scientist in the sense that you are where you really have to have a deep deep faith that what you're doing matters and that down the road you will see it's you know it's it's application um unless you unless it's just the beauty of it you know for its own sake compels you and draws you in but i'm ultimately a very kind of pragmatic person and so i and i'm also impatient so i need to see that a lot a lot sooner but how about you i'm sure you have many thoughts on this i'd love to hear them well yes very interesting and as we were kind of joking about before like everyone especially in an age of specialization abundance of specialisms that being able to like connect across them is really important like having not had first-hand experience in for example any of the stories presented and that's like the microcosm of the discussion with frameworks and people's lives kind of meeting from the ground up so patience that's an interesting thing and you point into like how different people seemingly display behaviorally different amounts of patience and they probably experience it differently too but also it's patience for different things right intermittent fasting over this many hours or this many days all the things might align for one person at one time and it very easy to be patient behaviorally versus not versus some other strategy and so trying to think about that in an active inference way it's like in active inference the question is always about what is the decision right now or inference and action so how should i be perceiving and learning and acting in the moment given the long horizon or the the stated horizon so what is going to be the best move in this chess move given the kind of uncertainty about the future of the game include including like the uncertainty about what the other side will do or something like that or even uncertainty over one's own actions and then patience is like well what is happening when the best action selection is to not engage with something so what is it that actually values for something that is valuable like it's easy to be patient for something that's just a

purely negative experience right i'll just i'll hold off not interested but being patient we kind of are applying to something where it's like i do want to call this person before the time i said i would call them but i'm getting impatient or like i want this research project to be over i wish that this was faster which everyone involved in research or maybe analogously like it would be awesome if myself or if this patients were seemingly on their recovery faster or something how does it come to be that sometimes we wait in the moment and other other times we don't and like is it okay or when should we do stuff yeah gosh so much there so let's start with your question how how does it come to be how how do we i mean i quote you in the book saying that the world is uh century rich and causal poor and i i just love that quote because it wraps up in this tight little package the the the problem with highly reinforcing drugs and behaviors which is that in the moment they feel like the right thing and they feel good but the long-term consequences of repeated use is to get us to a potentially very dark place but we don't necessarily see the connection between those two things right so how is it that we come to see that connection and i think as a psychiatrist having listened to thousands and thousands of people's narrative life stories over time that one of the ways that we get there is actually the autobiographical self narrative that that that words are tools and that those tools are a way not just to organize past experience but actually create road maps for the future so it's not just um you know the the way that we tell our stories about our lives are very very impactful i think puko or somebody like that said said that that autobiographical narrative is the only way that we have to document live time which is really really interesting like for all of the cool technology that we have the only thing that documents live time is storytelling so and i've seen this again and again when my patients come in and they tell an autobiographical narrative in which they are constantly the victim of the world and other people and that's why they're addicted those people are not going to get better and part of what i need to do over time is to move them toward what i consider to be a more truthful narrative which is that there's always something that we have contributed to the problem you know maybe we just showed up and that was our contribution but it's never always entirely somebody else's fault and for people who get into a sustained and robust recovery you can hear the difference in their autobiographical narratives and one of the fundamental differences they're able to tolerate the shame that comes with acknowledging their character defects and the mistakes that they made and what they contributed to their messed up situation which is just basically closer to the truth so telling the truth is really important telling truthful stories is really important and i think that those stories actually become the tool by which we are able to remain patient in the

face of you know delayed gratification that's what keeps us able to not to do something that we should do in the moment but don't feel like doing or to abstain from doing something that we feel like doing in the moment but shouldn't do um and you know whether that story takes a framework based on neuroscience or whether there's a religious or spiritual framework like um you know my reward will come in another life there are many different ways to you know build that story uh but that story has got to be the thing that allows that person to kind of you know delay this gratification in the service of living a more flourishing life i'd like to ask a follow-up question from the chat and then give a remark neuropixel nice screen name asks how can we learn to build patients for healthy habits okay how can we learn to build that's so so good so i think there are a couple uh there are a couple stages first um it's to recognize that when you first stop doing a bad habit you will suffer more so that the initial your initial experience will be increased pain not like immediate reward from stopping that bad habit so i think so that like setting up expectations in that way but that the longer you can go abstaining from that bad habit the the faster you will get to that place where you begin to feel better and you don't miss that bad habit in fact you you start to reap the benefits remember you have to know it's going to get worse before it gets better everybody talks about practicing mindfulness and what that is but actually there's almost no better opportunity for practicing mindfulness than to give up something that's bad for you that you really like a bad habit because what will happen is it'll stir up all kinds it's very disruptive and i think then the ability to learn that you are not your thoughts and emotions but that you can stand some distance from that and not respond to that but just objectively observe those things is is really important i think do it giving up a bad hat bad habit with other people so that you're not doing it alone is really important um because you can then you know go to other people and and sort of suffer together and then i think meaning making is really important so to give up the bad habit so that you yourself can feel better but also what is the what is the greater meaning of abstaining uh from that thing and often it's a consuming process and you know a greater meaning could be saving the planet right i'm doing this for me but it's also good for the world thanks for the answer so yeah one of those words you mentioned was expectations which is a core idea in bayesian statistics and a long history of like bayesian brain predictive coding predictive processing many different approaches to neuroscience behavior and so on and active inference does build on that and uses the idea of expectations as far as expected futures and things like that a lot and so it's interesting to think about the role of dopamine and like you said set up for this because then it'll basically go that way and you'll stay on track whereas if you expect to go

faster or different you're going to feel like worse or anxious or uncertain about how to proceed and it's totally modeled in an active inference way by thinking about like the estimate is not how many points how many are am i getting it's how many points relative to how many points i expect and then one can be doing better or worse or as expected and if things are going well there's positive valence if things are going as expected it's like neutral valence and then things are going poor there's like anxiety and negative valence right and so that is one way where moving from like a reward only perspective to be like well yes there's an expectation of reward and precision along that path will be pleasurable or neutral but it's not going to be like winning winning winning it'll be like neutral with oscillation around neutral right yeah and i can tell you you know in the like the lived experience of treating patients this is absolutely key so one of the things that we do in our clinic um is to help people get off of chronic opioids like you know oxycontin or vicodin or norco or chronic benzodiazepines like xanax thallium atom and you would think this is not that complicated but actually a lot of patients are looking for help with this and can't find it and they find it in our clinic and the one thing that i would say we provide which is the real value is to establish appropriate expectations because what happens is they'll go see a doctor and they'll say i need to get off a xanax or i really want to get off the xanax and the doctor says oh that's easy you know just go down by you know uh you know 50 of your starting dose over the next week and you'll be fine and the patients do that or same thing with their opioid they do that and they're in horrific psychological and physical pain because they're in withdrawal or maybe they even have a seizure right their experience is totally worse or much worse than what their doctor predicted for the doctor is like the respected authority so then they feel angry they also feel betrayed they also feel like they're weird like why why am i they come to us and my patients have told me again and again doctor argue the most important thing you said to me when we went down by my opioid or my benzo by a little bit and we go very slowly is that i would feel worse before i feel better and i really do i way emphasize that so that if they go and they taper and it's not even it's not as bad as what i then they're like happy it's like you know you told me it was going to be bad for a while until my body really recruited it re-equilibrated and it was bad but it wasn't like i could do it you know so that setting expectations it's really really important like in the real world it's almost like there's a wanting for reduction or sobriety but actually they don't like it behaviorally not on the psychological what are they thinking what does the survey say but actually like well you don't like the green room if you aren't spending your time in it it's how we would say it about the other kinds of animals so it's kind of like interesting like where do

we like something but not want it what is dopamine and other neurotransmitters doing then how about the situations where we do or don't have patience like those are such great questions and entry points for our own life and for research and that's like kind of the bridge is the introduction of the paper or the context for the grant is like kind of could be that bridge but then sometimes it is more about how it appears rather than how it is yeah and i think the variable of time is so important um for all of these things like if i mean you know we can sign up for a lot of suffering if you know it's time limited right like okay i can do this because dr lemke said i would feel like crap for two weeks and i would start to feel better and so you know you can hang in there for that or or whatever you know the the intervention is that that makes a huge difference returning to the quote on page 75 about uh the world is sensory rich and causal poor you know personal communication uh when 2018 because i think there's a very interesting difference or something and how i'm thinking about it now oh interesting okay i'd love to hear that so i think that was when i was just beginning to learn about active inference and free energy principle and bayesian brain and i was thinking well wow it's not just like signal detection on the pixels or whatever there's the causal inference and then that's like you followed up like we know the donut tastes good in the moment but we are less aware that eating a donut every day for a month adds five pounds to our waistline we get that understanding through personal experience relationships science research we build that understanding that expectation of cause in and then all of a sudden our behavior is like more natural to switch in the moment so i was thinking you know starry eyed graduate student sensory rich in causal poor i think now i might say sensory poor and causal poor because the sensory part is of extremely limited frequency only some chemicals that we can detect only some sounds that we hear even within a domain we're not always paying attention our blind spot is evidence that we're in a generative model of vision because the sensory data is sparse and noisy and blurry and so it's like in the world that we can finally disengage from this like watching a movie perspective and it's like it's so generative it's so much about our expectations of the cause and also we're not just watching a movie a vision and we're expecting expecting and generating even that yeah so powerful i think this is really why i i think there's so much power in our self-narratives because i think those self-narratives that we project onto the world absolutely impact what we perceive i just yesterday i was meeting with a very beloved uh patient of mine who happens to be you know a a a graduate student um you know who has had a really mixed experience in his lab and feels like he hasn't many friends and nobody likes him also feels like um like they're all jerks and they're kind of out to get him and what i tried

to get him to see is that he has a narrative and a vision that he superimposes on the world such that all he remembers or appreciates is like the slights right and he doesn't he doesn't have access to the other things because he's so entrenched in this particular version of the world just to give like a personal example um you know i talk about in the book my comp very complicated relationship with my mom and how part of the healing for my relationship was to get out of this place where i'm always saying how how terrible she is and instead appreciating really the great things about her which are many and also you know what i've contributed to the problem but before i could do i was engaging in behaviors that just reinforced my worldview and i'll give you a very tangible example um one of the things that that you know had bothered me about her was that i would email her and she wouldn't respond to my email in a timely or what i felt was appropriate fashion and this came up at one point where i emailed about her about something that was really important i felt it was really important and she didn't respond and i gave her a week and then i gave her a month and i gave her three months and i got angrier and angrier as time went on and you know railed about it to my husband say see this is exactly what i'm talking about and then you know i was eventually just able to just forget about it and whatever move on and then i was on email and kind of looking through i don't know kind of cleaning up my email and discovered that my original email that i had sent to her that i was so angry that she hadn't responded to was actually in my draft box it had never been sent and i think there are so many ways in which we do that where where we don't even get the opportunity to see it like it was just in this case i did but we we sort of do these things to sort of you know hold on to this world new which typically has other people being the bad actors and it doesn't force us to look at ourselves thanks so one interesting piece there is like we might be totally on board for the past being a reconstruction in the present the generative aspect of memory and then it's also easy to believe in the generative aspect of the future it hasn't happened yet we don't know about it and then as per that predictive coding type of like sensory generative model we're also now casting so it's like the past the present the future you know the three areas that we know about they're all generative so there is reception of signals but also like all of those are generative and then you pointed to like this idea of kind of a cognitive degree of freedom in our narrative and so we might even model that as like an action or a movement in a narrative space and so how do we make appropriate actions in narrative space because do we just say okay graduate student no they're awesome or how do we know what to say and how to actually yeah so so this is a student of mine that i mentioned this patient of mine who happens to be a graduate student so he's about to you know finish up

and go on to a postdoc and i actually suggested that to him as sort of a as an experiment right because i often talk about this as an experiment i said you know as you move on to your next lab why don't you try an experiment where you just take as like an a priori supposition that everybody in the lab loves you that that that they want the best for you that they are your friends and anytime you have like a paranoid thought or a negative thought about them really critically appraise that and really even doubt that and fact check it or reality test it with somebody else that you know and i think through that he might be able through iterative processes because it's such a feed-forward cycle the problem is when we're paranoid or suspicious of other people they sense it and they weigh back off and then as soon as they back off we get more paranoid as we should because they've given us the signal that they've backed up and then that you know it's a reinforcing cycle so if we kind of discard that primary narrative and then act in the world as if right not and not in a fake way but as if certain things were true that we didn't necessarily believe up front and i have to ask my patients to do this all the time especially when my patients who are addicted and whose lives are being destroyed by their addiction i say to them you have to imagine and essentially have faith giving up your drug of choice will lead to a better life right now what you see is two choices you see continuing to use and or stop using and being miserable and i'm holding out to you the possibility of stop using and being happy or being happier than you are while you're using but it is a leap of faith because you have to are going to have to wait a long time and do a lot of things where really you don't feel that at all you just have to believe that it's true and approach every day as if that is true the other thing i talk about in my book which is maybe a little bit of a paradox given what i just said is this idea of radical honesty and how important truth-telling is to this process again because truth-telling is important to build true narratives and true narratives are important to be able to know how to make decisions going forward in our lives and it's a small thing but you know my patients will say and patients in recovery will say i can't lie about anything when i was using i of course lied about my drug use but i also lied about stuff that it didn't even matter and my favorite anecdote is my patient who's now in really sustained recovery doing great he says you know when i was using i was in the lying habit if i was at mcdonald's and a patient called and said hey where are you i'd say hey i'm a burger king if i was a bird king my patient my friend would call it say yeah i'm in mcdonald's so it got to be like this work like like lies about everything so part of his recovery and this is what i actually prescribe to patients is radical truth telling like you cannot tell a lie about anything because that telling the truth i think helps us first of all it's it's cognitively effortful

but secondly it really grounds us in what is true and real and helps us in an iterative fashion stay closer to that to build these truthful narratives which i think are really important so there's a bit of a paradox there because i'm saying on the one hand pretend stuff that you don't feel is but which might be true which you can't see but at the same time be really honest about like your daily actions nice here here were some of the active themes that i kind of heard there action as a self-fulfilling prophecy and action as being in the business of acquiring evidence for a given generative model that's a big theme and also in order to switch action policies there has to be a transient suppression of precision on the policy like if we're sitting down we have to transiently implicitly or explicitly believe that it could be a different way otherwise we'll just continue to get evidence that we're sitting down and then um it's a very active way to kind of escape that gaslight paradox you were talking about setting a confident prior for the next epoch and so that just today you're low on how you believe others think of you and uncertain or you're low and precise but great it's actually all in the past now behaviorally and then in your next endeavor you could just try starting higher in an unknown future but just peg it higher and try to be more precise which is going to look like being more skeptical looking for evidence on the upside and so it's like instead of focusing on the past and trying to re-narrativize the past we can include the behavior of the past which for sure is inalterable into a prospective narrative of the future that makes actions like patience maybe more plausible yeah love it um i mean dislikes love that science speak it's so exciting and it maps so well to like you know these kinds of aphorisms or sort of the approach that we take in psychiatry and psychotherapy using very different language and that's always really exciting and it just reminds me so much of you know the alcoholics anonymous saying of taking it one day at a time which is so fundamental for this whole process because what happens when we take it one day at a time it means that we wake up and we say the past is the past the future is the future all i need to do today is focus on what i can do today to make my life and your life and our lives a little bit better without putting too heavy on me like i mean i'm not going to go and like save people in ukraine obviously but what are the things that i can do i can pick up that piece of garbage so it kind of narrows the time scope it forces us to be in the here and now which also expands our senses so that we can take in more actual data so that i'm not like ruminating and cogitating on my own you know sort of self-involved you know problems which is where like you know most of us spend a lot of our headspace i don't know why we do that but we do but instead i noticed that there's like a reese's peanut butter cup wrapper on the ground i could pick that up right that's one thing that or i i could notice that i have the urge to

lie about why i'm late for this meeting but i could not i could choose not to do that i can choose that today that's kind of exciting like to me that sort of opens up all kinds of possibilities because i'm not future catastrophizing i'm also not future grandiosing you know where they like oh i have to do this i have to do that or i'm not going to be good enough i'm not ruminating on my pastels i'm saying you know what today is the only unit of time that i really need to focus on and a string i always say to my patients a string of good days makes a string of good weeks and a string of good weeks it's a string of good months and before you know it you've got a couple good decades and that's a lifetime and that's a life well lived which is what we all want and to even bring in one other active term the affordance which is the capacity for action which comes from the ecological psychology field the affordance is relational and so it's about like the motivation by that entity and the agency that they feel they do or don't have from a narrative perspective and the affordances that the niche scaffolds and provides for them or not and also the narratives that the niche provides or not and then like that is what is going to be engaged in that behavioral decision in the moment and then you also talk about like adding friction as like a pragmatic strategy that i'm sure everyone has experimented with variously on different facets but it's like it's an example the pragmatic but also they're connected to behavioral and cognitive modeling in this perhaps even formal way exactly um well we covered many fun topics like just a few that i kind of wrote down where patients i guess the patients who you clinically see and having patience in the moment uh expectations action and taking action as a first perturbation then the dopamine as reward epistemic and pragmatic value and there's a lot of interesting research on active inference and dopamine signaling and then a lot of active terms like action priors confidence precision narrative information like all the amazing terms and just very fun and also powerful book and a lot of great material thank you daniel this has been really fun as i knew it would be as talking to you is always illuminating and surprising in fun and positive way i always learn then i have so much respect for your you know intelligence and creativity it's always exciting thank you and hope to talk to you and any others from the farm in the future sounds great bye see ya you